A Constructive Introduction to Quantum Field Theory

Lecture 6: Φ^4_3 on the torus II. A priori estimates, UV stability and the limiting FBSDE

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Abstract

We prove the central estimate of the construction: bounds, uniform in the lattice, on the error Z of the approximate forecast and on the time the forecast spends at large fields. The two halves of the FBSDE of Lecture 5 give two inequalities, neither of which closes alone. The forward equation becomes, through the field equation of Lecture 4, a cubic elliptic equation at each resolution, with no memory: its noise is the filtered measurement noise, sub-Gaussian uniformly in the interaction. This equation is coercive: testing it bounds the weighted large-field action, scale by scale, by the tails of the noise and by the backward energy, and the maximum principle gives a pointwise bound driven by the cube. The backward equation is linear in Z: a conditional representation formula bounds the backward energy by an integrable bounded-field source, a boundary term, and a small multiple of the large-field action. The small coefficient is $C\gamma^2/q_0$, independent of the lattice and of the moment order, so one choice of q_0 closes the loop. The clipping error of Lecture 5 is controlled exactly in this way. We give complete proofs in the scalar (one-site) model, where all the mechanisms are visible with explicit constants, and indicate the changes needed on the lattice.

We then close the argument. At a fixed lattice the error of the approximate forecast vanishes at infinite resolution, because the explicit action matches the bare action there; this removes the boundary term of the energy estimate, and gives bounds uniform in the lattice (UV stability). The order of limits is essential: first the upper scale at fixed lattice, then the lattice. The uniform energy bounds give compactness of the forecast in space and scale, by a maximum principle and a linear elliptic estimate, and tightness of the lattice measures in H^{-s}, $s > \frac{1}{2}$. Identifying the limit requires two further arguments: weak convergence of the deterministic coefficients against compact resolvents, and strong compactness of the backward integrand, which follows from the positivity of the posterior covariance and its exact Riccati equation. The result is a solution of the FBSDE on the continuum torus, driven by a Brownian motion obtained as the limit of the lattice innovations, with no additional noise, whose terminal law is a subsequential limit of the lattice measures and satisfies a renormalised integration by parts formula. Tilting the filter, as in the argument of Hairer and Steele, gives uniform exponential moments of the tested field beyond the Gaussian order. We also give the Dyson–Schwinger equations along the scales, and discuss uniqueness in finite volume through the dynamics. We end with what is not proved and with the infinite-volume problem.

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Introduction

At the end of Lecture 5 we had, at each lattice cutoff N, the exact system (Theorem 10.6)

$$\begin{aligned} \mathrm{d}Y_q &= [b_q(Y_q) + H_q(Y_q)\, Z_q]\mathrm{d}q + H_q(Y_q)\,\mathrm{d}W_q, \\ \mathrm{d}Z_q &= -[b_q(Y_q) + H_q(Y_q)\, Z_q]\mathrm{d}q + \Xi_q\,\mathrm{d}W_q, \qquad Y_{q_0} = Z_{q_0} = 0, \end{aligned} \tag{1}$$

where Y is the approximate forecast built from the explicit action $\bar{\Gamma}_q$, $Z = \psi - Y$ is its error, and $\psi = Y + Z$ is the exact forecast, a martingale. The inputs from Lecture 5 are:

A) *Global coercivity* (Lemma 12.1): $H_q(y)$ is comparable to $(K_q + \gamma y^2)^{-1}$ and $\|H_q(y)\|_{p\to p} \le (1+\varepsilon_q)/q$, $\varepsilon_q = C(g_q + g_q^2)$.

B) *The source* (Proposition 12.2): $b = b^{\mathrm{b}} + b^{\mathrm{l}}$ with $\|b_q^{\mathrm{b}}\|_\infty \le C\gamma^{7/2} q^{-5/2}$ and $\|b_q^{\mathrm{l}}(y)\|_2^2 \le C\gamma^4 q^{-5} \|t_q(y)\|_6^6$, where $t_q(y) = y 1_{|y|>r_q}$.

C) *The coordinate equation* (12.5): $K_q(Y_q + \varrho_q(Y_q)) + \frac{\gamma}{3} Y_q^3 = K_q \Phi_q = P_q$, with a small Lipschitz correction, $\|D\varrho_q\|_{p\to p} \le \varepsilon_q$.

The goal is to show that Z is small and that Y rarely visits large fields, *uniformly in* N. This is UV stability. Neither equation can be estimated alone: the forward equation contains Z, and the source of the backward equation is controlled by the large fields of Y.

How a priori estimates for FBSDEs are usually obtained. For an FBSDE with Lipschitz coefficients one estimates $\mathbb{E}\|Z_q\|^2$ by Itô's formula and Gronwall, and closes either on a short time interval or under a monotonicity condition [MY99]. Here the time interval is infinite, the coefficients are not uniformly Lipschitz in the relevant norms, and the coupling is not small in a Lipschitz sense. What we have instead is structure:

- the backward equation is *linear* in Z, given Y, so it can be solved by a conditional representation formula with an explicit propagator;
- the forward equation is, by the field equation of Lecture 4, a cubic *elliptic* equation at each resolution, with no memory, which is coercive and can be estimated pathwise;
- its noise is a conditional expectation of the measurement noise, hence sub-Gaussian whatever the interaction;
- the two estimates are linked through two weighted energies whose weights are matched.

Sections 13–15 prove these estimates at a fixed lattice and a finite upper scale. Sections 16–18 remove the cutoffs and identify the limit.

13 The energies

13.1 Moments at fixed cutoff

All estimates below are first proved on a finite interval $[q_0, T]$ and at fixed N. There, everything is finite.

Lemma 13.1. (Preliminary moments) *At fixed N and T, the maxima over $[q_0, T]$ of P, Y, ψ, Z have all moments, and so does $\int_{q_0}^T (\|\mathrm{Cov}_q\|_{\mathrm{HS}}^2 + \|\Xi_q\|_{\mathrm{HS}}^2)\mathrm{d}q$. Consequently the energies E_T, J_T and G_T defined below have all moments at fixed N, T.*

Proof. P has all maximal moments, since ν_N has all moments. Global convexity, $D^2\bar{\Psi}_q \ge c\,q$ and $D\bar{\Psi}_q(0) = 0$, gives $\|Y_q\| \le C\|P_q\|/q$. Doob's inequality for the martingale $\psi_q = \mathbb{E}_q \Phi_\infty$ gives its maximal moments, and $Z = \psi - Y$. For the integrands, $\|\mathrm{Cov}_q\|_{\mathrm{HS}} \le \mathrm{Tr}\mathrm{Cov}_q \le \mathbb{E}_q\|\Phi_\infty\|^2$, and $\|H_q(Y_q)\|_{\mathrm{HS}} \le C\sqrt{\dim H_N}/q$. The energies involve, besides Y and Z, the field ζ_q of Section 13.2 and $\varrho_q(Y_q)$; finite-dimensional norm equivalence and $\|\varrho_q(y)\| \le \varepsilon_q\|y\|$ give their moments. □

These constants depend on N and T. They are used only to justify the absorptions below, never in the final bounds.

13.2 The field equation

Lecture 5 realises the process as a measurement: by (10.5), $P_q = (q-q_0)\,\Phi_\infty + B_{q-q_0}$, with B a Brownian motion. Since $\psi_q = \mathbb{E}_q \Phi_\infty$, the noise seen at resolution q is explicit:

$$\eta_q := P_q - (q-q_0)\,\psi_q = \mathbb{E}_q[B_{q-q_0}]. \tag{13.1}$$

This is the noise observable of Theorem B.1 of Lecture 4 in the profile (10.2), where $S_q = C_q$ and the regulator is $q-q_0$. Its bound (Proposition B.3 of Lecture 4) is here one line.

Lemma 13.2. (The noise) *For every convex Ψ on H_N, $\mathbb{E}\,\Psi(\eta_q) \le \mathbb{E}\,\Psi(B_{q-q_0})$, whatever the interaction. In particular $\zeta_q := C_q\,\eta_q$ satisfies, for all x and $u \in \mathbb{R}$,*

$$\mathbb{E}\,\mathrm{e}^{u\zeta_q(x)} \le \exp\left(\frac{u^2}{2}\,(q-q_0)\,a^2(q)\right), \qquad (q-q_0)\,a^2(q) \le C\sqrt{q}, \tag{13.2}$$

uniformly in N.

Proof. Conditional Jensen in (13.1), with $\Psi(f) = \mathrm{e}^{u\langle C_q\delta_x, f\rangle}$, and (10.3). □

So ζ_q has the tails of the free field at resolution q. It is not Gaussian, and it depends on the interaction through ψ, but its bounds do not. Insert $P_q = \eta_q + (q-q_0)\,(Y_q + Z_q)$ into the coordinate equation (C), and use $K_q = K_{q_0} + q - q_0$. With

$$V_q := Y_q + \varrho_q(Y_q) - \zeta_q, \tag{13.3}$$

(V_q is not to be confused with the potentials V_t, V_N of Lectures 4–5) we obtain the *field equation*

$$K_{q_0} V_q + \frac{\gamma}{3}\,Y_q^3 = (q-q_0)\,(\zeta_q - \varrho_q(Y_q) + Z_q), \qquad Y_q = V_q + \zeta_q - \varrho_q(Y_q). \tag{13.4}$$

This is the equation we estimate: the field equation (B.4) of Lecture 4 in these coordinates. Three remarks:

- it is an elliptic equation at each resolution, with *no memory*: the past enters only through the noise η_q;
- the only trace of the backward equation is $(q-q_0)\,Z_q$, at the same resolution;
- the identity is pathwise: no expectation has been taken.

The mass of (13.4) is only q_0, so at large resolution all the coercivity must come from the cube. The next section shows that it does.

13.3 Thresholds and energies

Define the signed excess of V over three quarters of the threshold, the large-field part of Y, and the large values of the noise,

$$w_q = \mathrm{sgn}(V_q)\,\Big(|V_q| - \tfrac{3}{4}r_q\Big)_+, \qquad t_q = Y_q\,1_{|Y_q|>r_q}, \qquad \xi_q = \zeta_q\,1_{|\zeta_q|>r_q/32}, \tag{13.5}$$

all pointwise in space, with $r_q = (q/\gamma)^{1/2}$. The three random energies are

$$\begin{aligned} E_T &= \frac{1}{\gamma}\int_{q_0}^T q^{1/2}\|Z_q\|_2^2\,\mathrm{d}q && \text{(backward energy)},\\ J_T &= \int_{q_0}^T q^{-3/2}\,[q\,\|w_q\|_4^4 + \gamma\,\|w_q\|_6^6 + \gamma\,\|t_q\|_6^6]\,\mathrm{d}q && \text{(large-field action)},\\ G_T &= \int_{q_0}^T [q^{-1/2}\,\|\xi_q\|_4^4 + \gamma\,q^{-3/2}\,\|\xi_q\|_6^6]\,\mathrm{d}q && \text{(noise tails)}. \end{aligned} \tag{13.6}$$

Only *thresholded* sixth powers are integrated over all scales: $\mathbb{E}\|\zeta_q\|_6^6$ can be of order $q^{3/2}$, so an unthresholded noise contribution to J would not be integrable uniformly in N. The weights are matched to each other and to the source bounds:

quantity	weight	reason
$\|Z_q\|_2^2$	$q^{1/2}$	forcing: $q^{-3/2}\cdot\gamma^{-1}q^2\|Z\|_2^2$
$\gamma\|t_q\|_6^6$	$q^{-3/2}$	source: $q^{7/2}\|b^1\|^2\le C\gamma^3\cdot\gamma q^{-3/2}\|t\|_6^6$
$\|b_q^1\|_2^2$	$q^{7/2}$	backward propagator s/q and $\int_q^\infty s^{-3/2}\mathrm{d}s=2q^{-1/2}$

Exercises

Exercise 13.1. (**Matched weights**) Check the three weights in the table. Show from (13.4) that the forward estimate, with the weight $q^{-3/2}$ on the large-field action, needs the weight $q^{1/2}$ on $\|Z\|_2^2$. Show that the backward estimate of Section 15 also holds with weight $q^{1/2+\sigma}$ for $0<\sigma<\frac12$. Which term breaks at $\sigma=\frac12$, and how does the feedback coefficient depend on σ?

Exercise 13.2. (**Moments of the noise**) Show from Lemma 13.2 that $\mathbb{E}\|\zeta_q\|_6^6\le 15\,((q-q_0)\,a^2(q))^3\le C\,q^{3/2}$ and $\mathbb{E}\|\xi_q\|_6^6\le C\,q^{3/2}\exp\,(-c\sqrt{q}\,/\gamma)$. For $V_N=0$, compute the covariance of ζ_q and compare it with the bound. On which modes is the bound sharp?

Exercise 13.3. (**The memory form**) Let $X_q=\int_{q_0}^q C_s\,\mathrm{d}W_s$, so that $K_q X_q=W_q+\int_{q_0}^q X_s\,\mathrm{d}s$, and $V_q'=Y_q+\varrho_q(Y_q)-X_q$. Using $P_q=W_q+\int_{q_0}^q\psi_s\,\mathrm{d}s$, derive the equation with memory

$$K_q V_q'+\frac{\gamma}{3}Y_q^3=\int_{q_0}^q (V_s'-\varrho_s(Y_s)+Z_s)\,\mathrm{d}s.$$

Here the noise is exactly Gaussian, but the memory has the size of the mass term. Why would testing it with w^3 need the Hardy inequality $\|q^{-1}\int_0^q f\|_{L^4(q^{-1/2}\mathrm{d}q)}\le\frac{8}{7}\|f\|_{L^4(q^{-1/2}\mathrm{d}q)}$, and where would the margin come from? What does (13.4) trade to avoid this?

14 Forward coercivity

14.1 The scalar model

On one site the Laplacian disappears. We also drop the small Lipschitz correction ϱ (restored in Proposition 14.3) and treat the noise and the error as given numbers. Then (13.4) is a scalar equation at each q, and all the constants can be checked by hand.

Lemma 14.1. (**Scalar forward coercivity**) *Let $\gamma,q>0$, $a\ge0$, $0\le p\le q$ and $v,g,z\in\mathbb{R}$ with*

$$a\,v+\frac{\gamma}{3}(v+g)^3=p\,(g+z). \tag{14.1}$$

Let $r=(q/\gamma)^{1/2}$, $w=\mathrm{sgn}(v)\,(|v|-\frac34 r)_+$, $\xi=g\,1_{|g|>r/32}$. Then

$$\frac{q}{8}w^4+\frac{\gamma}{16}w^6\le 54\,q\,\xi^4+\frac{4225}{6}\gamma\,\xi^6+\frac{12}{\gamma}q^2z^2. \tag{14.2}$$

Moreover $t=(v+g)\,1_{|v+g|>r}$ satisfies $|t|\le\frac{32}{7}\,(|w|+|\xi|)$.

Proof. Multiply (14.1) by w^3, where $w\neq0$; there $|v|=|w|+\frac34 r$ and v,w have the same sign, so $a\,v\,w^3\ge0$. Since $|g|\le\frac{r}{32}+|\xi|$, $\mathrm{sgn}(w)\,(v+g)\ge|w|+\frac{11}{16}r-|\xi|$ (we give up $\frac{1}{32}r$ to leave room, on the lattice, for ϱ). The cube being increasing, (14.3) and $\gamma r^2=q$ give

$$\frac{\gamma}{3}w^3\,(v+g)^3\ge\frac{\gamma}{12}w^6+\frac{q}{4}w^4+\frac{q\,r}{13}|w|^3-\frac{4225}{6}\gamma\,\xi^6.$$

This is the large-field coercivity: above the threshold the cube provides a sixth power, a quartic term of the size of a mass q, and a cubic term which dominates the noise below the threshold. On the right side, $p\,g\,w^3 \le \frac{qr}{32}|w|^3 + q|\xi||w|^3$, and the first term is absorbed. By Young's inequality, $q|\xi||w|^3 \le \frac{q}{8}w^4 + 54\,q\,\xi^4$ and $q|z||w|^3 \le \frac{\gamma}{48}w^6 + \frac{12}{\gamma}q^2 z^2$. This gives (14.2). For the last claim, $|v+g| \le |w| + \frac{3}{4}r + \frac{r}{32} + |\xi|$, so on $\{|v+g|>r\}$ we have $\frac{7}{32}|v+g| \le |v+g| - \frac{25}{32}r \le |w| + |\xi|$. □

Lemma 14.2. (Two elementary inequalities) *For $x, \rho \ge 0$, $(x-\rho)^3 \ge \frac{3}{4}x^3 - 65\rho^3$. For $u, \rho, r \ge 0$,*

$$\frac{1}{3}u^3\left(u + \frac{11}{16}r - \rho\right)^3 \ge \frac{1}{12}u^6 + \frac{1}{4}r^2u^4 + \frac{1}{13}r^3u^3 - \frac{4225}{6}\rho^6. \tag{14.3}$$

Proof. $\frac{1}{4}x^3 + 64\rho^3 - 3x^2\rho = \frac{1}{4}(x-8\rho)^2(x+4\rho) \ge 0$, so $(x-\rho)^3 \ge x^3 - 3x^2\rho - \rho^3 \ge \frac{3}{4}x^3 - 65\rho^3$. Apply it with $x = u+b$, $b = \frac{11}{16}r$: $\frac{1}{3}u^3(x-\rho)^3 \ge \frac{1}{4}u^3x^3 - \frac{65}{3}u^3\rho^3$, and $x^3 \ge u^3 + 3b^2u + b^3$ with $\frac{3}{4}b^2 = \frac{363}{1024}r^2 > \frac{1}{4}r^2$ and $\frac{1}{4}b^3 = \frac{1331}{16384}r^3 > \frac{1}{13}r^3$. Finally $\frac{65}{3}u^3\rho^3 \le \frac{1}{6}u^6 + \frac{4225}{6}\rho^6$, and $\frac{1}{4} - \frac{1}{6} = \frac{1}{12}$. □

What the scalar lemma says. The excess of the forecast over the threshold is controlled, at each resolution separately, by two things only: how large the noise is, whose tails are known and exponentially small at large q, and the error Z. There is no smallness in the coupling: the estimate is uniform in γ once the threshold scales like $r_q = (q/\gamma)^{1/2}$. It is pathwise, and it needs no integration in q.

14.2 The lattice version

Proposition 14.3. (Forward coercivity) *For q_0 large (depending on γ), at every cutoff N, pathwise and for every q,*

$$q\|w_q\|_4^4 + \gamma\|w_q\|_6^6 + \gamma\|t_q\|_6^6 \le C\,(q\|\xi_q\|_4^4 + \gamma\|\xi_q\|_6^6 + \gamma^{-1}q^2\|Z_q\|_2^2). \tag{14.4}$$

Consequently, for every T,

$$J_T \le C\,(G_T + E_T), \qquad \|G_T\|_{L^p(\mathbb{P})} \le C\gamma^3 p^5 \quad (p \ge 1), \tag{14.5}$$

with C independent of N, T and p.

Proof: the changes from the scalar case. Test (13.4) with w_q^3 in $L^2(\Lambda_N)$ and proceed as in Lemma 14.1, with Hölder's inequality in space. Three new terms appear.

The elliptic term. $\langle w^3, K_{q_0}V\rangle \ge 0$. Indeed $w^3 = \phi'(V)$ for the convex function $\phi(v) = \frac{1}{4}(|v| - \frac{3}{4}r)_+^4$, $Vw^3 \ge 0$, and for any nondecreasing ϕ',

$$\langle \phi'(V), -\Delta_N V\rangle = N^{-1}\sum_{x\sim x'} (\phi'(V(x)) - \phi'(V(x')))\,(V(x) - V(x')) \ge 0,$$

summing over unordered nearest-neighbour pairs. So the elliptic term only helps.

The coordinate correction. Split $\varrho_q(y) = \varrho_q^0 + \varrho_q^1$ according to $|y| \le r_q$ or not (writing $\varrho_q(y) = L_q(y)\,y$ with L the averaged derivative). Then $\|\varrho_q^0\|_\infty \le \varepsilon_q r_q \le r_q/32$ shifts the threshold, and $\|\varrho_q^1\|_j \le \varepsilon_q\|t_q\|_j$, $j = 4, 6$. Then $|Y| \le |w| + \frac{13}{16}r_q + |\xi| + |\varrho^1|$; on $\{|Y| > r_q\}$ this gives $\frac{3}{16}|Y| \le |w| + |\xi| + |\varrho^1|$, so for small ε

$$\|t_q\|_j \le C\,(\|w_q\|_j + \|\xi_q\|_j), \qquad \|\varrho_q^1\|_j \le C\,\varepsilon_q\,(\|w_q\|_j + \|\xi_q\|_j).$$

This also supplies the t part of (14.4). These are operator bounds: ϱ is nonlocal, but small in every L^p. In the cube and on the right side of (13.4) the error is $\rho = |\xi| + |\varrho^1|$. The bounded parts on the right cost $\frac{qr}{16}\|w\|_3^3$, which is absorbed by $\frac{qr}{13}\|w\|_3^3$ from the cube. The cube error is at most $C\gamma\,(\|\xi\|_6^6 + \varepsilon^6\|w\|_6^6)$, and Young's inequality gives $Cq\,(\|\xi\|_4^4 + \varepsilon^4\|w\|_4^4)$. The w parts are absorbed for small ε: these are deterministic smallness conditions on g_{q_0}, independent of p.

The error. $|\langle w^3, (q-q_0)\,Z\rangle| \le \frac{\gamma}{48}\|w\|_6^6 + \frac{12}{\gamma}q^2\|Z\|_2^2$. Multiplying (14.4) by $q^{-3/2}$ and integrating gives $J_T \le C\,(G_T + E_T)$.

Tails of the noise. By Minkowski's inequality in x, for $j=4,6$, $\|\|\xi_q\|_j^j\|_{L^p(\mathbb{P})} \le \sup_x \|\|\xi_q(x)|^j\|_{L^p(\mathbb{P})}$, and by Hölder and (13.2), with threshold $r_q/32 = \frac{1}{32}(q/\gamma)^{1/2}$, this is at most

$$(\mathbb{E}|\zeta_q(x)|^{2jp})^{\frac{1}{2p}}\,\mathbb{P}(|\zeta_q(x)| > r_q/32)^{\frac{1}{2p}} \le C_j p^{j/2} q^{j/4} \mathrm{e}^{-c\sqrt{q}/(p\gamma)},$$

and integrating in q with $u = \sqrt{q}/(p\gamma)$ gives $\|G_T\|_p \le C\gamma^3 p^5$. □

No information on the interacting field has been used, apart from the equation itself and the posterior structure of the noise. The tails of ζ are those of the free field whatever the interaction: this replaces the subtraction of a Gaussian field.

14.3 A pointwise bound

The maximum principle gives more than the integrated estimate: a bound in L^∞ driven by the cube alone.

Proposition 14.4. (L^∞ coercivity) *For q_0 large, at every cutoff, pathwise and for every q,*

$$\|Y_q\|_\infty \le C\left[\left(\frac{q}{\gamma}\right)^{1/3}\|\zeta_q\|_\infty^{1/3} + \|\zeta_q\|_\infty + q^{1/4} + q\|Z_q\|_2\right]. \tag{14.6}$$

In particular there is $c>0$ such that $t_q = 0$, i.e. $\|Y_q\|_\infty \le r_q$, whenever $\|\zeta_q\|_\infty + q\|Z_q\|_2 \le c\,r_q$: large fields occur only when the noise or the error is large.

Proof. Let $p = q - q_0$, $d = \zeta_q - \varrho_q(Y_q)$, and remove the error by $\mathcal{H} = p\,C_{q_0}Z_q$. Then $\|\mathcal{H}\|_\infty \le p\,a(q_0)\|Z_q\|_2 \le C\,q\|Z_q\|_2$, since the kernel of C_{q_0} is in L^2 uniformly in N by (10.3). By (13.4), $W = V - \mathcal{H}$ and $D = \mathcal{H} + d$ satisfy

$$K_{q_0}W + \frac{\gamma}{3}(W+D)^3 = p\,d, \qquad Y_q = W + D.$$

At a positive maximum of W, $K_{q_0}W \ge 0$, so $W + D \le (3p\|d\|_\infty/\gamma)^{1/3}$ there; similarly at a negative minimum. Hence $\|Y_q\|_\infty \le 2\|D\|_\infty + (3q\|d\|_\infty/\gamma)^{1/3}$. With $\|\varrho_q(Y_q)\|_\infty \le \varepsilon_q\|Y_q\|_\infty$, the term $2\,\varepsilon_q\|Y_q\|_\infty$ is absorbed, and by Young's inequality $(q\,\varepsilon_q\|Y_q\|_\infty/\gamma)^{1/3} \le \frac{1}{4}\|Y_q\|_\infty + C\,(q\,\varepsilon_q/\gamma)^{1/2}$, where $q\,\varepsilon_q/\gamma \le C\sqrt{q}$ since $g_q = \gamma\,q^{-1/2}$. This proves (14.6). For the last claim use $q/\gamma = r_q^2$: the terms of (14.6) are at most $C\,(c^{1/3} + c + g_q^{1/2})\,r_q$, and $g_q^{1/2} \le g_{q_0}^{1/2}$ is small. □

Lemma 14.5. (Supremum of the noise) *Uniformly in N and in the interaction,*

$$\|\mathcal{N}\|_{L^p(\mathbb{P})} \le C\sqrt{p}, \qquad \mathcal{N} := \sup_{q\ge q_0} q^{-1/4}\,(\log(2+q))^{-1/2}\|\zeta_q\|_\infty.$$

Proof. At fixed q the norm $f \mapsto \|C_q f\|_\infty$ is convex, so by Lemma 13.2 $\|\zeta_q\|_\infty$ is dominated by the supremum of the Gaussian field $C_q B_{q-q_0}$. Its variance is at most $C\sqrt{q}$, and for $\theta < 1$ its increments have variance at most $C\sqrt{q}\,(\sqrt{q}|x-y|)^\theta$ uniformly in N, so Dudley's bound gives the mean $\lesssim q^{1/4}\sqrt{\log(2+q)}$ and Gaussian concentration gives fluctuations at the scale $q^{1/4}$. For the supremum over $s \in [Q, 2Q]$, write $C_s = C_{2Q}K_{2Q}C_s$, where $\|K_{2Q}C_s\|_{\infty\to\infty} \le 2$. Since $\mathrm{d}\eta_q = (I - (q-q_0)\mathrm{Cov}_q)\,\mathrm{d}W_q$ by (10.8), η is a martingale, $\|C_{2Q}\eta_s\|_\infty$ is a submartingale, and Doob's inequality gives, for $p \ge 2$, $\|\sup_{s\in[Q,2Q]}\|\zeta_s\|_\infty\|_{L^p} \le C\,Q^{1/4}\,(\sqrt{\log(2+Q)} + \sqrt{p})$. Markov's inequality with $p \asymp \lambda^2 \log(2+Q)$ bounds the probability that the normalised supremum over $[Q, 2Q]$ exceeds λ by $(2+Q)^{-c\lambda^2}$, for $\lambda \ge \lambda_0$. This is summable over dyadic Q, and $\|\mathcal{N}\|_p \le C\sqrt{p}$ follows. □

Exercises

Exercise 14.1. **(Without the cube)** Replace $\frac{\gamma}{3}(v+g)^3$ by 0 in (14.1). Show that $v = p\,a^{-1}(g+z)$, so that the estimate fails as $q\to\infty$ at fixed mass a. Which term of (14.2) replaces the missing mass, and how does the threshold $r=(q/\gamma)^{1/2}$ enter?

Exercise 14.2. **(Discrete convexity)** Prove that $\langle \phi'(u), -\Delta_N u\rangle \ge 0$ for every convex $\phi\in C^1$ and $u\in H_N$, and that equality holds iff $\phi'(u(x)) = \phi'(u(x'))$ on every edge $x\sim x'$ with $u(x)\neq u(x')$.

Exercise 14.3. **(Numerical check)** Solve (14.1) numerically for random γ, q, a, p, g, z, and check (14.2). Take g Gaussian with variance $\sqrt{q}$, $z=0$, $\gamma=1$, and compare $|v+g|$ with $(q|g|/\gamma)^{1/3}$ and with r. Is the exponent $5/12$ of $(q|g|)^{1/3}\approx q^{5/12}$ attained on one site? What should improve it on the lattice?

15 Backward feedback and closure

15.1 The backward representation

The backward equation in (1) is linear in Z. Its propagator is the adapted operator solution of

$$\partial_s \mathcal{P}_{q,s} = \mathcal{P}_{q,s} H_s(Y_s), \qquad \mathcal{P}_{q,q} = I, \qquad s\ge q. \tag{15.1}$$

By (A), $\|\mathcal{P}_{q,s}\|_{2\to 2}\le \exp\int_q^s \frac{1+\varepsilon_u}{u}\mathrm{d}u \le L_0\,\frac{s}{q}$, with $L_0 = \exp\left(\int_{q_0}^\infty \varepsilon_u \frac{\mathrm{d}u}{u}\right) \le \exp\left(C\,(g_{q_0}+g_{q_0}^2)\right)$.

Lemma 15.1. **(Conditional representation)** *For $q_0\le q\le T$, at fixed N,*

$$Z_q = \mathbb{E}_q\Big[\,\mathcal{P}_{q,T} Z_T + \int_q^T \mathcal{P}_{q,s}\, b_s(Y_s)\,\mathrm{d}s\Big]. \tag{15.2}$$

Proof. By the product rule and (1), $\mathrm{d}_s(\mathcal{P}_{q,s}Z_s) = \mathcal{P}_{q,s}H_sZ_s\mathrm{d}s - \mathcal{P}_{q,s}(b_s+H_sZ_s)\mathrm{d}s + \mathcal{P}_{q,s}\Xi_s\mathrm{d}W_s = -\mathcal{P}_{q,s}\,b_s\mathrm{d}s + \mathcal{P}_{q,s}\,\Xi_s\mathrm{d}W_s$. The stochastic integral is a true martingale by Lemma 13.1 and the deterministic bound on $\mathcal{P}$. Integrate from q to T and take $\mathbb{E}_q$. □

So Z is the expected source accumulated on the unresolved scales, transported back to q, plus the boundary error. This is the conditional-expectation form of a backward equation, as in $F_t = \mathbb{E}_t[F_T]$ for the force (Theorem 7.4); here the source is the defect.

15.2 Three contributions

Split $Z = Z^{\mathrm{t}} + Z^{\mathrm{b}} + Z^{\mathrm{l}}$ according to the terminal term and the two parts of the source. We estimate each in the scalar manner: norms of the propagator times norms of the source.

Bounded fields. By (B),

$$\|Z^{\mathrm{b}}_q\|_2 \le \int_q^\infty L_0\frac{s}{q}\,C\gamma^{7/2}s^{-5/2}\,\mathrm{d}s = 2L_0C\gamma^{7/2}q^{-3/2}, \qquad \frac{1}{\gamma}\int_{q_0}^\infty q^{1/2}\|Z^{\mathrm{b}}_q\|_2^2\,\mathrm{d}q \le C\gamma^6 q_0^{-3/2}.$$

This is where the two correction steps of Lecture 5 are used: with one step the source would be $O(s^{-2})$ and the first integral would diverge logarithmically.

Boundary. $q\|Z^{\mathrm{t}}_q\|_2 \le L_0\,\mathbb{E}_q\|T Z_T\|_2$, so, with $\delta_{N,T,p} := \|T Z_T\|^2_{L^{2p}(\mathbb{P};L^2)}$,

$$\left|\frac{1}{\gamma}\int_{q_0}^T q^{1/2}\|Z^{\mathrm{t}}_q\|_2^2\mathrm{d}q\right|_{L^p(\mathbb{P})} \le \frac{L_0^2}{\gamma}\int_{q_0}^\infty q^{-3/2}\mathrm{d}q\ \ \delta_{N,T,p} = \frac{2L_0^2}{\gamma\,q_0^{1/2}}\ \delta_{N,T,p}.$$

At fixed N, $\delta_{N,T,p}\to 0$ as $T\to\infty$ (Lemma 16.1).

Large fields. Let $A^{\mathrm{l}} := \int_{q_0}^T s^{7/2}\|b^{\mathrm{l}}_s(Y_s)\|_2^2\mathrm{d}s \le C\gamma^4\int s^{-3/2}\|t_s\|_6^6 \le C\gamma^3 J_T$ by (B). By Cauchy–Schwarz in s,

$$\|Z^{\mathrm{l}}_q\|_2 \le \frac{L_0}{q}\,\mathbb{E}_q\int_q^T s\|b^{\mathrm{l}}_s\|_2\mathrm{d}s \le \frac{L_0}{q}\Big(\int_q^\infty s^{-3/2}\mathrm{d}s\Big)^{1/2}\mathbb{E}_q[(A^{\mathrm{l}})^{1/2}],$$

so that, by Jensen's inequality, $\|Z^1_q\|_2^2 \le 2L_0^2 q^{-5/2}\,\mathbb{E}_q A^1$. Integrating, and using that $\mathbb{E}_q$ is a contraction on $L^p(\mathbb{P})$,

$$\left\| \frac{1}{\gamma}\int_{q_0}^{T} q^{1/2}\|Z^1_q\|_2^2 \mathrm{d}q \right\|_{L^p} \le \frac{2L_0^2}{\gamma}\int_{q_0}^{\infty} q^{-2}\mathrm{d}q\,\|A^1\|_{L^p} \le \frac{C\gamma^2}{q_0}\|J_T\|_{L^p}.$$

This is how the clipping error is controlled. The large-field source b^1 is not small; but its squared L^2 norm is controlled by the sixth power of the large-field part, $\|b^1_q\|_2^2 \le C\gamma^4 q^{-5}\|t_q\|_6^6$ (B), and after transport by the propagator and integration with the energy weight it contributes to the backward energy only through J_T, with the small factor γ^2/q_0.

Proposition 15.2. (Backward feedback) *At every cutoff,*

$$\|E_T\|_{L^p} \le C\gamma^6 q_0^{-3/2} + C\gamma^{-1} q_0^{-1/2}\,\delta_{N,T,p} + C\gamma^2 q_0^{-1}\|J_T\|_{L^p}, \tag{15.3}$$

with C independent of N, T and p.

Proof. $\|Z\|^2 \le 3\,(\|Z^{\mathrm{t}}\|^2 + \|Z^{\mathrm{b}}\|^2 + \|Z^{\mathrm{l}}\|^2)$ and the three estimates above. □

15.3 Closing the loop

Theorem 15.3. (Cutoff-uniform energy estimate) *There is q_0, depending only on γ, such that for every N, T and $p \ge 1$,*

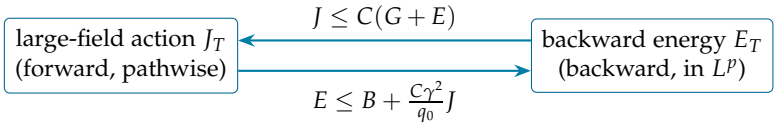
$$\|E_T\|_{L^p} + \|J_T\|_{L^p} \le C_\gamma\,(1 + p^5 + \delta_{N,T,p}). \tag{15.4}$$

Proof. Let $x = \|E_T\|_p$ and $y = \|J_T\|_p$. By Propositions 14.3 and 15.2,

$$y \le A_0\,(C\gamma^3 p^5 + x), \qquad x \le B_0 + B_1\,\delta_{N,T,p} + \eta\,y, \qquad \eta = \frac{C\gamma^2}{q_0},$$

with A_0, B_0, B_1 and the constant in η independent of N, T, p, and bounded for all q_0 beyond the thresholds already required. Choose q_0 such that also $A_0\eta \le \frac{1}{2}$. Then $y \le A_0(C\gamma^3 p^5 + B_0 + B_1\,\delta_{N,T,p}) + \frac{1}{2}y$. Since $y < \infty$ by Lemma 13.1, it can be absorbed, and x follows. □

The structure of the argument is the point:



Forward coercivity estimates the large fields using the backward energy; the backward representation returns that energy with a small coefficient. Neither half closes on its own. The smallness is in the *feedback*, γ^2/q_0, not in the coupling of either equation, and it is independent of the moment order, so a single q_0 closes the loop for every p. The growth p^5 is then inherited from the tails of the noise, G_T, and gives stretched exponential moments $\mathbb{E}\exp\{c\,(E_\infty + J_\infty)^{1/5}\} < \infty$ once the boundary term is removed. Note that $\gamma^2/q_0 = g^2_{q_0}$.

15.4 Consequences

The boundary term $\delta_{N,T,p}$ is removed in Section 16, at each fixed N, by letting $T \to \infty$. Assuming this for the moment, we obtain

$$\sup_N \|E^N_\infty\|_{L^p} + \sup_N \|J^N_\infty\|_{L^p} \le C_\gamma\,(1 + p^5), \qquad \sup_N \mathbb{E}\|Z^N_q\|_2^2 \le C_\gamma q^{-5/2}. \tag{15.5}$$

The second bound follows from the representation formula on $[q,\infty)$: the three contributions are, up to the factor 3, $O(q^{-3})$, 0 and, by the computation above, $2L_0^2 q^{-5/2}\,\mathbb{E}\,A^1$. So the approximate forecast differs from the exact one by $O(q^{-5/4})$ in $L^2(\mathbb{P};L^2)$ (root mean square), uniformly in the lattice. Since $\|H_q\|\le C/q$, the coupling term $H_q Z_q$ in the forward equation is $O(q^{-9/4})$ in $L^2(\mathbb{P};L^2)$, integrable in q: the forward equation is, up to an integrable drift, the explicit SDE $\mathrm{d}Y = b(Y)\,\mathrm{d}q + H(Y)\,\mathrm{d}W$. The same computation holds along the path. At $T=\infty$ the boundary part vanishes (Lemma 16.1) and, with A^1 now integrated up to ∞ (it has all moments, since $A^1\le C\gamma^3 J_\infty$), $q^{5/2}\|Z_q\|_2^2 \le 3\,(C\gamma^7 q_0^{-1/2} + 2L_0^2\,\mathbb{E}_q A^1)$. Since $\mathbb{E}_q A^1$ is a martingale, Doob's inequality gives, for $p\ge 2$,

$$\Big\| \sup_{q\ge q_0} q^{5/2}\|Z_q\|_2^2 \Big\|_{L^p(\mathbb{P})} \le C_\gamma\,(1+p^5). \tag{15.6}$$

With Proposition 14.4 and Lemma 14.5 this gives pointwise bounds on Y for all resolutions, used in Section 17.

Remark 15.4. (**Comparison with the variational method**) In the Boué–Dupuis approach (Theorem 9.1) one estimates a drift u which is the costate up to $\dot{Q}^{1/2}$, and the renormalisation is a change of variables $u = -4\lambda\dot{Q}^{1/2}{:}X^3{:}+r$ (in the normalisation $\lambda\int{:}\varphi^4{:}$ of Lecture 4) [BG20]. Here the explicit action $\bar{\Gamma}$ plays the role of all such changes of variables at once, in the forecast coordinate, and Z plays the role of the remainder r. The coercivity of the quartic term is used in both, but here it is used *pathwise in the forward equation*, while the remainder is estimated in expectation through the backward equation.

Exercises

Exercise 15.1. (**The scalar backward estimate**) On one site, with $H_q(y) = (q+\gamma y^2)^{-1}$, show that

$$\mathcal{D}_{q,s} = \frac{s}{q}\exp\left(-\int_q^s \frac{\gamma Y_u^2}{u\,(u+\gamma Y_u^2)}\mathrm{d}u\right) \le \frac{s}{q}.$$

In which sense do large fields make the backward equation more contractive? Redo the three estimates of Section 15 on one site.

Exercise 15.2. (**Absorption needs finiteness**) Give an example of nonnegative numbers x,y (possibly infinite) with $y\le 1+x$, $x\le 1+\frac{1}{2}y$ and $y=\infty$. Where is Lemma 13.1 used in the proof of Theorem 15.3?

Exercise 15.3. (**Moment growth**) From $\|E_\infty + J_\infty\|_p \le C\,(1+p^5)$ for all $p\ge 1$, deduce $\mathbb{E}\exp\{c\,(E_\infty+J_\infty)^{1/5}\} < \infty$ for small c.

Removing the cutoffs

Sections 13–15 proved, at every lattice cutoff N and every upper scale T,

$$\|E_T\|_{L^p} + \|J_T\|_{L^p} \le C_\gamma\,(1+p^5+\delta_{N,T,p}), \qquad \delta_{N,T,p} = \|T Z_T\|^2_{L^{2p}(\mathbb{P};L^2)}, \tag{2}$$

for the backward energy E_T and the large-field action J_T of (13.6). Three things remain.

i. Show that the boundary term vanishes: $TZ_T \to 0$ as $T\to\infty$ at fixed N (Section 16).

ii. Turn the uniform energy bounds into compactness of the processes and of the measures (Section 17).

iii. Identify the limit: the deterministic coefficients, the noise, and the equations (Section 18).

The structure of (iii) is instructive: convergence of the fields alone does not suffice to pass to the limit in an FBSDE, because the backward martingale might converge to something with *extra noise*. Here the statistical meaning of the backward variable, a posterior mean with a positive covariance, saves us.

16 The boundary at infinite resolution

16.1 The boundary lemma

At a fixed lattice, the terminal resolution is $q=\infty$, and the field at resolution q, $\Phi_q = C_q P_q$, converges to the terminal field Φ_∞. The approximate action $\bar\Gamma_q$ was built to converge to the bare action: $\mu_q \to \mu_N$, $\Theta_q \to 0$, and the nonlinear corrections were integrated from $q=\infty$, so that they vanish there. This is the boundary condition of the backward equation.

Lemma 16.1. (Boundary lemma) *At fixed N, for every $p<\infty$, as $q\to\infty$,*

$$\|Y_q-\Phi_\infty\|_{L^p(\mathbb{P};H_N)} + \|q\,Z_q\|_{L^p(\mathbb{P};H_N)} \le C_{N,p}\,q^{-1/2}, \qquad \textit{hence} \qquad \delta_{N,T,p} \le C_{N,p}\,T^{-1}. \tag{16.1}$$

The constants depend on N; this is not a uniform estimate.

Proof. All norms on H_N are equivalent, and we do not track the dependence on N.

The field at resolution q. By (10.5), $\Phi_q = C_q P_q = (q-q_0)\,C_q\,\Phi_\infty + C_q B_{q-q_0}$, and $(q-q_0)\,C_q - I = -C_q K_{q_0} = O(q^{-1})$, $\|C_q B_{q-q_0}\|_{L^p} = O(q^{-1/2})$. So $\|\Phi_q-\Phi_\infty\|_{L^p} \le C_{N,p}\,q^{-1/2}$.

The approximate forecast. Global convexity gives $\|Y_q\| \le C\|P_q\|/q$, bounded in every L^p uniformly in q. The coordinate equation $\Phi_q = Y_q + C_q D\bar\Gamma_q(Y_q)$ and the bounds $|\mu_q| \le C_N$, $\|\Theta_q\|_{\mathrm{row}} = O_N(q^{-1/2})$ and $\|D\tilde h_q(y)\|_\infty \le C\mathsf{S}_4(h_q^{[2]})\|y\|_\infty^3 \le C\gamma^2 q^{-1/2}\|y\|_\infty^3$ (Taylor from the fourth order, since the low jets vanish) show that $D\bar\Gamma_q(Y_q)$ is bounded in L^p, so $\|Y_q-\Phi_q\|_{L^p} = O(q^{-1})$. Moreover $|\mu_q-\mu_N| = O_N(q^{-1})$, and therefore

$$\|D\bar\Gamma_q(Y_q) - D\,V_N(\Phi_\infty)\|_{L^p} \le C_{N,p}\,q^{-1/2}. \tag{16.2}$$

Thus *the explicit force at the approximate forecast converges to the bare force at the terminal field.*

The error. Subtract the approximate coordinate identity $P_q = K_q Y_q + D\bar\Gamma_q(Y_q)$ from the exact posterior identity (10.10), $P_q = K_q\psi_q + \mathbb{E}_q[D\,V_N(\Phi_\infty)]$:

$$q\,Z_q = q\,C_q\,[D\bar\Gamma_q(Y_q) - \mathbb{E}_q D\,V_N(\Phi_\infty)]. \tag{16.3}$$

Let $G = D\,V_N(\Phi_\infty)$ and $G_q^0 = D\,V_N(\Phi_q)$, which is $\mathcal{F}_q$-measurable. Then $\|\mathbb{E}_q G - G\|_{L^p} \le \|\mathbb{E}_q(G-G_q^0)\|_{L^p} + \|G_q^0-G\|_{L^p} \le 2\|G-G_q^0\|_{L^p} = O(q^{-1/2})$, by the cubic Lipschitz bound for $D\,V_N$ and the first step. With (16.2) and $\|q\,C_q\| \le 1$ this proves the claim. □

The proof uses no estimate uniform in N, and no concentration of the posterior: only that the observed estimator $\Phi_q = C_q P_q$, which is $\mathcal{F}_q$-measurable, converges to the hidden field. In Lecture 4 language, the forecast and the field at resolution q have the same limit at infinite resolution, and the force $F_q = K_q(\psi_q - \Phi_q)$ converges to the bare force $-D\,V_N(\Phi_\infty)$ (Exercise 16.1; recall $F_q = \mathbb{E}_q[-D\,V_N(\Phi_\infty)]$ by (10.9), so this is the conditional-expectation step of the proof).

16.2 UV stability

Now let $T\to\infty$ at fixed N in (2): the paths and the law ν_N do not depend on T, E_T and J_T increase to E_∞ and J_∞, and $\delta_{N,T,p}\to 0$ by Lemma 16.1. Only then take the supremum over N.

Theorem 16.2. (**UV stability [FBSDE26]**) *For q_0 large enough, depending only on γ,*

$$\sup_N (\|E_\infty^N\|_{L^p} + \|J_\infty^N\|_{L^p}) \le C_\gamma (1+p^5), \qquad \sup_N \mathbb{E}\|Z_q^N\|_2^2 \le C_\gamma q^{-5/2}, \tag{16.4}$$

and $\sup_N \mathbb{E}\exp\{c\,(E_\infty^N + J_\infty^N)^{1/5}\} < \infty$ for small $c > 0$.

Proof. Monotone convergence in (2), then the representation formula (15.2) on $[q,\infty)$, whose terminal term vanishes by Lemma 16.1. The exponential moment follows from the polynomial growth by Markov's inequality with $p \asymp R^{1/5}$. □

The order of limits matters. The constant $C_{N,p}$ in Lemma 16.1 diverges with N, and we never take a supremum over N of the remainder $C_{N,p}/T$. The infinite-resolution boundary is used only qualitatively, at each lattice separately.

Exercises

Exercise 16.1. (**The force at infinite resolution**) Show that, at fixed N, $F_q = K_q(\psi_q - \Phi_q) \to -DV_N(\Phi_\infty)$ in every L^p, and that $F_q = K_q Z_q - D\bar{\Gamma}_q(Y_q)$. Interpret the last identity: the force observable of Lecture 4 is the explicit force at the approximate forecast, corrected by $K_q Z_q$.

Exercise 16.2. (**Why the stronger boundary**) Using the bound on the boundary term in the proof of Proposition 15.2, explain why $Z_T \to 0$ alone would not make it vanish, and why the rate $\|TZ_T\| = o(1)$ of Lemma 16.1 is what is needed.

17 Compactness

17.1 One Hilbert space

To compare lattices, partition $\mathbb{T}^3$ into cubic cells $\square_x$ of side N^{-1} centred at $x \in \Lambda_N$, and let

$$(\iota_N u)(z) = u(x)\ (z \in \square_x), \qquad (j_N f)(x) = N^3 \int_{\square_x} f, \qquad \mathsf{P}_N = \iota_N j_N.$$

$\iota_N : H_N \to H = L^2(\mathbb{T}^3)$ is an isometry, $j_N = \iota_N^*$, and $\mathsf{P}_N \to I$ strongly. Fields are embedded by ι_N, operators by $\iota_N \cdot j_N$. The embedded innovation has covariance P_N; in the limit it becomes a cylindrical Brownian motion on H, obtained as the limit of the lattice innovations.

17.2 Spatial suprema by a maximum principle

The energy bounds control L^4 and L^6 norms of large fields. Spatial suprema at finite resolution come from the L^∞ coercivity of Proposition 14.4.

Lemma 17.1. (**Supremum after closure**) *Let $\mathcal{Y} = \sup_{q \ge q_0} q^{-5/12} (\log(2+q))^{-1/6} \|Y_q\|_\infty$ and $\mathcal{Z} = \sup_{q \ge q_0} q^{5/4} \|Z_q\|_2$. For q_0 large, with $\mathcal{N}$ the supremum of the noise of Lemma 14.5,*

$$\mathcal{Y} \le C_\gamma (1 + \mathcal{N} + \mathcal{Z}), \qquad \|\mathcal{N}\|_{L^p} \le C\sqrt{p}, \qquad \|\mathcal{Z}^2\|_{L^p} \le C_\gamma (1+p^5), \tag{17.1}$$

uniformly in N.

Proof. By the definitions, $\|\zeta_q\|_\infty \le q^{1/4} (\log(2+q))^{1/2} \mathcal{N}$ and $q\|Z_q\|_2 \le q^{-1/4} \mathcal{Z}$. Inserting these in (14.6), the first term gives $\gamma^{-1/3} q^{5/12} (\log(2+q))^{1/6} \mathcal{N}^{1/3}$, and the others are of lower order. The bound on $\mathcal{Z}$ is (15.6), whose boundary term vanishes by Lemma 16.1. □

Any exponent below $\frac{1}{2}$ keeps Y below the threshold r_q at large q, and we only need moments of the supremum of $\|Y_q\|_\infty$ on finite intervals of scales. The exponent $5/12$ comes from the cube alone. The memory form of Exercise 13.3, with a weight in q absorbing the memory, gives every exponent above $\frac{1}{4}$, at the price of that absorption.

In particular *finite-resolution* spatial suprema of Y have all moments uniformly in N. At the terminal resolution there is no such bound: Φ_∞ is a distribution.

17.3 Regularity in space and scale

On a finite scale interval $I=[q_0,Q]$ write the field equation (13.4) as the linear elliptic equation

$$K_{q_0} V_q = f_q, \qquad f_q = (q-q_0)\,(\zeta_q - \varrho_q(Y_q) + Z_q) - \frac{\gamma}{3} Y_q^3. \tag{17.2}$$

The energies control $\mathbb{E}\int_I \|Y_q\|_6^6 \mathrm{d}q$ (since $|Y|^6 \le r_q^6 + |t_q|^6$) and $\mathbb{E}\int_I \|Z_q\|_2^2 \mathrm{d}q$, and $\mathbb{E}\|\zeta_q\|_2^2 \le C\sqrt{q}$. Hence $\mathbb{E}\int_I \|f_q\|_2^2 \le C_I$, and testing with V,

$$\mathbb{E}\int_I (\|\nabla_N V_q\|_2^2 + \|V_q\|_2^2)\,\mathrm{d}q \le C_I.$$

So the part V of the approximate forecast is in H^1, and $Y = V + \zeta - \varrho(Y)$ inherits the regularity of the noise ζ. This is that of the free field, whatever the interaction: by Lemma 13.2, $\mathbb{E}\|\zeta_q\|_{H^\sigma}^2 \le C\,(q-q_0)\sum_k (1+|k|^2)^\sigma\,(q+c|k|^2)^{-2}$, finite uniformly in N for $\sigma < \frac{1}{2}$.

Proposition 17.2. (Forward compactness) *For $0<\sigma,\theta<\frac{1}{2}$,*

$$\sup_N \mathbb{E}\Big[\int_I \|\iota_N Y_q^N\|_{H^\sigma}^2 \mathrm{d}q + \|\iota_N Y^N\|_{H^\theta(I;H)}^2\Big] < \infty.$$

The embedded approximate forecasts are tight in every space–time L^p, and polynomials of them converge along strongly convergent subsequences. The forward equation gives the Hölder bound $\sup_N \mathbb{E}\,\|\iota_N Y^N\|_{C^{1/4}(I;H)}^8 < \infty$, and averaging in time against the H^σ bound gives uniform Fourier tails $\lim_m \sup_N \mathbb{E}\sup_{q\in I}\|(I-\mathsf{F}_m)\,\iota_N Y_q^N\|_2^2 = 0$ (F_m the projection on frequencies $|k|\le m$); hence they are also tight in $C(I;H)$.

The spatial bound combines the H^1 bound on V, the H^σ regularity of ζ ($\sigma<\frac{1}{2}$, for which the cell embedding is uniformly bounded), and the contraction property of ϱ: since ϱ commutes with lattice translations τ_a, (12.5) gives $\|\tau_a\varrho(Y) - \varrho(Y)\|_2 \le \varepsilon_q \|\tau_a Y - Y\|_2$, so the increments of Y are controlled by those of $V+\zeta$; the scale regularity uses the forward equation, whose drift $b(Y)+H(Y)Z$ has bounded integrated square by Proposition 12.2 and the energies, and whose martingale part has bounded bracket density. The supremum bound of Lemma 17.1 upgrades L^2 compactness to every L^p.

17.4 The terminal field

The approximate forecast converges at infinite resolution in negative Sobolev spaces, uniformly in N (to the same limit as the forecast ψ). The martingale part of Y has bracket $\int H^2$, and the positive inequality $H_q^2 \le \frac{C}{q} C_q$ gives

$$\int_Q^\infty \|(1-\Delta)^{-s/2} H_q\|_{\mathrm{HS}}^2 \mathrm{d}q \le C\int_Q^\infty \frac{1}{q}\sum_k \frac{(1+|k|^2)^{-s}}{q+c|k|^2}\,\mathrm{d}q \le C_s\, Q^{-(s-\frac{1}{2})}, \qquad \tfrac{1}{2}<s<\tfrac{3}{2}, \tag{17.3}$$

where $q+c|k|^2$ bounds the lattice symbol from below (aliases of the cell embedding only increase the frequency). The drifts are integrable at infinity in H: the bounded-field drift is $O(q^{-5/2})$, the large-field drift is controlled by $\int q^{7/2}\|b^1\|^2 \le C\gamma^3 J_\infty$ and Cauchy–Schwarz, and $\int \|H_q Z_q\|\mathrm{d}q \le C\int q^{-1}\|Z_q\| \le C\gamma^{1/2}E_\infty^{1/2}$. More precisely, the squared drift tails beyond Q are $O(Q^{-3})$ (bounded-field), $O(Q^{-5/2})\gamma^3 J_\infty$ (large-field) and $O(Q^{-3/2})\gamma E_\infty$ (the HZ term), all at most $Q^{-(s-\frac12)}$ for $s<\frac32$. By Doob's inequality, for $\frac12<s<\frac32$,

$$\sup_N \mathbb{E}\sup_{t\ge Q}\|Y_t - Y_Q\|_{H^{-s}}^2 \le C_s\, Q^{-(s-\frac12)}. \tag{17.4}$$

Letting $t\to\infty$ at fixed N (Lemma 16.1) this holds with Y_t replaced by Φ_∞. Together with the tightness at finite resolution, this gives:

Corollary 17.3. (Tightness of the lattice measures) *For every $s>\frac12$ the embedded laws $\nu_N\circ\iota_N^{-1}$ are tight in $H^{-s}(\mathbb{T}^3)$, and $\sup_N \mathbb{E}\,\|\iota_N\Phi_\infty^N\|_{H^{-s}}^p<\infty$ for every p.*

For the moments: the drift integrals have all moments by (16.4), and the H^{-s} martingale has a deterministic bracket bound by (17.3) at $Q=q_0$, so the Burkholder–Davis–Gundy inequality gives all moments.

The threshold $s>\frac12$ is that of the free field in $d=3$: the interacting field is no rougher than the Gaussian one, and at every finite resolution its part V is smoother (H^1).

17.5 Exponential integrability: the Hairer–Steele argument

Theorem 16.2 gives only stretched exponential moments of the energies. Hairer and Steele [HSt22] found a very simple argument for much stronger integrability of the field. Let W be a functional with $\mathcal{Z}=\mathbb{E}\mathrm{e}^W<\infty$, and let $\hat\nu=\mathcal{Z}^{-1}\mathrm{e}^W\nu$ be the tilted measure. For every set A, $\hat\nu(A)\le\mathcal{Z}^{-1}\exp(\sup_A W)\nu(A)$, so

$$\mathcal{Z}\le\frac{\exp(\sup_A W)}{\hat\nu(A)}. \tag{17.5}$$

To bound the exponential moment it is enough to know that the tilted measure gives mass $\frac12$ to some fixed set, uniformly in the cutoffs. No moment of e^W has to be estimated. In [HSt22], $W=\frac{\beta}{4}\langle\varphi,\psi\rangle^4$, the tilted measure is formally invariant for the Φ^4_3 dynamics with the added force $\beta\psi\langle\Phi,\psi\rangle^3$, which is dominated by the cube, and the bound comes from coming down from infinity [MW17, MW20], uniformly in the initial condition. We now run the same idea with the estimates of this lecture: the role of the dynamics is played by the filter, which is available for every prior.

The filter of a tilted measure. Fix N and an even C^1 functional W such that DW has polynomial growth and $W\le c\|\varphi\|_4^4+C$ with $c<\lambda/4$. Then $\hat\nu_N\propto\mathrm{e}^W\nu_N$ has all moments. Observe a field $\hat\Phi\sim\hat\nu_N$ through the same channel, $P_q=(q-q_0)\hat\Phi+B_{q-q_0}$ with B independent of $\hat\Phi$, let $\hat\psi_q=\mathbb{E}_q\hat\Phi$, and let W also denote its innovation Brownian motion. **Keep the action $\bar\Gamma_q$ unchanged**: $Y_q=D\bar{\mathcal{K}}_q(P_q)$, $\hat Z_q=\hat\psi_q-Y_q$, and define $\hat E$, $\hat J$ as in (13.6) with these fields.

Proposition 17.4. (What survives a tilt) *With q_0 and the constants of Theorem 15.3, which do not depend on W:*

a) The pair $(Y,\hat Z)$ solves (1) with the same b, H. Lemmas 13.2 and 14.5 and Proposition 14.3 hold unchanged.

b) The estimate (15.4) holds with $\hat\delta_{N,T,p}=\|T\hat Z_T\|^2_{L^{2p}(\mathbb{P};L^2)}$.

c) At fixed N, $T\hat Z_T\to DW(\hat\Phi)$ in every L^p as $T\to\infty$. Therefore

$$\|\hat E_\infty\|_{L^p}+\|\hat J_\infty\|_{L^p}\le C_\gamma(1+p^5+\|DW(\hat\Phi)\|^2_{L^{2p}(\mathbb{P};L^2)}). \tag{17.6}$$

Proof. (a) Theorem 10.6 of Lecture 5 uses only $\mathrm{d}P = \hat{\psi}\,\mathrm{d}q + \mathrm{d}W$ and that $\hat{\psi}$ is a martingale: the coefficients come from $\bar{\Gamma}$ alone. The identity $\eta_q = \mathbb{E}_q B_{q-q_0}$ holds for every prior, and the rest of the forward estimate is pathwise, given $\hat{Z}$. (b) Lemma 13.1 needs only that $\hat{\nu}_N$ has all moments, and then Lemma 15.1, Proposition 15.2 and the absorption are unchanged. (c) As in Lemma 16.1: the posterior identity now reads $P_q = K_q\hat{\psi}_q + \mathbb{E}_q[D\,V_N(\hat{\Phi}) - D\,W(\hat{\Phi})]$, so (16.3) becomes $q\hat{Z}_q = q\,C_q[D\bar{\Gamma}_q(Y_q) - \mathbb{E}_q D\,V_N(\hat{\Phi}) + \mathbb{E}_q D\,W(\hat{\Phi})]$. The first two terms cancel in the limit as before, and $\mathbb{E}_q D\,W(\hat{\Phi}) \to D\,W(\hat{\Phi})$ in every L^p by martingale convergence, since $\hat{\Phi}$ is measurable for $\mathcal{F}_\infty$. Finally let $T \to \infty$ in (15.4), by monotone convergence. □

So the only trace of the tilt in the system is the terminal value of the error: the matching of $\bar{\Gamma}$ with the bare action now fails by exactly $D\,W$. To close, we must bound the size of the field by $\hat{J}$. This needs no information on $\hat{Z}$: the coordinate equation at resolution q contains the terminal field directly.

Lemma 17.5. (The terminal field and the large-field action) *Let ψ be smooth, $\psi_N = j_N\psi$, $X = \langle\hat{\Phi}, \psi_N\rangle$, and $\mathcal{N}_\psi = \sup_{q\ge 2q_0} q^{-1}|\langle B_{q-q_0}, \psi_N\rangle|$, which has Gaussian tails independently of the prior. There is C_1, depending only on $\|\psi\|_{C^2}$, such that for every prior, pathwise,*

$$|X|^5 \le C_2 + (4C_1\mathcal{N}_\psi)^5 + 2\gamma^{-1/2}(8C_1)^5\hat{J}_\infty, \qquad C_2 = (8C_1 r_{4q_0})^5. \tag{17.7}$$

Proof. Test the coordinate equation (12.5) with ψ_N, and use $P_q = (q-q_0)\hat{\Phi} + B_{q-q_0}$. For $q \ge 2q_0$ we have $q - q_0 \ge q/2$, $\|K_q\psi_N\|_2 \le C\,q\|\psi\|_{C^2}$ and $\|\varrho_q(Y)\|_2 \le \|Y\|_2$, so

$$|X| \le C\|\psi\|_{C^2}(\|Y_q\|_2 + \frac{\gamma}{q}\|Y_q\|_3^3) + 2\mathcal{N}_\psi.$$

Since $|Y| \le r_q + |t_q|$ and $\gamma r_q^3/q = r_q$, this gives, for a suitable C_1,

$$|X| \le C_1(r_q + \|t_q\|_6 + \frac{\gamma}{q}\|t_q\|_6^3 + \mathcal{N}_\psi). \tag{17.8}$$

Let $R = |X|$ and suppose that $R \ge 8C_1 r_{4q_0}$ and $R \ge 4C_1\mathcal{N}_\psi$. Choose Q with $r_Q = R/(8C_1)$, so that $Q \ge 4q_0$. For $q \in [Q/2, Q]$, (17.8) gives $\|t_q\|_6 + \frac{\gamma}{q}\|t_q\|_6^3 \ge R/(2C_1)$. In either case $\|t_q\|_6 \ge R/(8C_1)$, the second one because $\gamma/Q = (8C_1/R)^2$. Therefore

$$\hat{J}_\infty \ge \gamma\int_{Q/2}^{Q} q^{-3/2}\|t_q\|_6^6\,\mathrm{d}q \ge \frac{\gamma}{2}Q^{-1/2}(\frac{R}{8C_1})^6 = \frac{\gamma^{1/2}}{2}(\frac{R}{8C_1})^5.$$

This is (17.7). The choice $Q \sim \gamma R^2$ is the one in which the field of size R is exactly at its threshold r_Q: the scale adapts to the level, as the time does in coming down from infinity. □

Theorem 17.6. (Exponential integrability) *For q_0 as in Theorem 16.2, smooth ψ and $s > \frac{1}{2}$, there are $\beta > 0$ and $C < \infty$ such that, uniformly in N,*

$$\mathbb{E}\exp(\beta|\langle\iota_N\Phi_\infty^N, \psi\rangle|^{7/2}) \le C, \quad \mathbb{E}\exp(\beta\|\iota_N\Phi_\infty^N\|_{H^{-s}}^2) \le C. \tag{17.9}$$

The same bounds hold for every limit point ν of the lattice measures.

Proof. First bound. Take $W(\varphi) = \beta(1 + \langle\varphi, \psi_N\rangle^2)^{7/4}$. Then $\|D\,W\|_2^2 \le C\beta^2\|\psi\|_2^2(1 + |X|^5)$. Inserting (17.7) in (17.6) with $p = 1$,

$$\mathbb{E}\hat{J}_\infty \le C_\gamma(2 + C\beta^2\|\psi\|_2^2(1 + C_2 + C\mathbb{E}\mathcal{N}_\psi^5 + C\gamma^{-1/2}\mathbb{E}\hat{J}_\infty)).$$

At fixed N, $\mathbb{E}\hat{J}_\infty$ is finite by (17.6), since $\hat{\nu}_N$ has all moments. So for small β it can be absorbed, and then $\mathbb{E}\hat{J}_\infty$ and, by (17.7), $\hat{\mathbb{E}}|X|^5$ are bounded by a constant independent of N. By Markov's inequality, $\hat{\nu}_N(|X| \le K) \ge \frac{1}{2}$ for a fixed K, and (17.5) gives $\mathcal{Z}_N \le 2\exp(\beta(1 + K^2)^{7/4})$.

Second bound. Take $W(\varphi)=\frac{\beta}{2}\|\iota_N\varphi\|^2_{H^{-s}}$. Now $DW=\beta j_N(1-\Delta)^{-s}\iota_N\varphi$ and $\|DW\|_2\le\beta\|\iota_N\varphi\|_{H^{-s}}$. As $Y_{q_0}=0$ and $Y_q\to\hat\Phi$, integrating the forward equation and using (17.3) for the martingale (with $s<\frac{3}{2}$, which is enough by monotonicity of the norms) and Cauchy–Schwarz in q for the drifts as in the proof of (17.4),

$$\hat{\mathbb{E}}\|\iota_N\hat\Phi\|^2_{H^{-s}}\le C_s(1+\gamma^3\mathbb{E}\hat J_\infty+\gamma\mathbb{E}\hat E_\infty).$$

Thus (17.6) with $p=1$ closes again for small β, and we conclude as before. The limit points follow by Fatou's lemma and weak convergence. □

Remark 17.7. (How far this goes) The field equation at a fixed scale would only give $|X|^2\lesssim 1+\hat J$, hence Gaussian tails. Adapting the scale to the level improves this to the exponent $\frac{7}{2}$. The quartic exponent, which is expected to be optimal because it is the exponent of the formal density, would need (17.7) with $|X|^6$. This is out of reach of (17.6) as it stands: on one site, the saddle $qy+\lambda y^3=(q-q_0)X$ gives $\hat J_\infty\sim\gamma^{1/2}|X|^5$ (Exercise 17.2), so (17.7) cannot be improved, while $\|DW\|^2\sim\beta^2|X|^6$ for the quartic tilt. What is lost is in the backward estimate: the bound $\|\mathcal{D}_{q,s}\|\le L_0 s/q$ ignores that $H_q(Y)$ is much smaller than $1/q$ in the direction of a large field, much as the cube keeps the force $\beta\psi\langle\Phi,\psi\rangle^3$ under control in [HSt22]. On one site a computation with the sharp propagator of Exercise 15.1 suggests that the quartic tilt would then close. Under the identification of Section 18.6, the quartic bound for ν follows from the dynamics (Corollary 18.9).

Already (17.9) has two consequences. The moments of $\langle\Phi,\psi\rangle$ grow slowly enough for Carleman's condition, so the Schwinger functions of ν determine ν. And $\langle\Phi,\psi\rangle$ is not a nondegenerate Gaussian variable, since such a variable has no finite exponential moment of order $\frac{7}{2}$; that $\langle\Phi,\psi\rangle$ is nondegenerate is not proved here.

Exercises

Exercise 17.1. (One site: the sharp threshold) On one site, with $\nu\propto\exp(-\frac{\lambda}{4}x^4-\frac{q_0}{2}x^2)$, show that $\mathbb{E}\exp(\frac{\beta}{4}x^4)$ is finite if and only if $\beta\le\lambda$. Which step of the proof of Theorem 17.6 requires β small, and how small?

Exercise 17.2. (The saddle) On one site let y_q solve $qy+\lambda y^3=(q-q_0)X$ for large X. Show that $y_q\approx(qX/\lambda)^{1/3}$ for $q\ll\gamma X^2$, and $y_q\approx X$ beyond. Compute $\gamma\int q^{-3/2}t_q^6\,\mathrm{d}q$ and show that it is of order $\gamma^{1/2}X^5$, dominated by $q\sim\gamma X^2$. Why is the adaptive choice of Q in Lemma 17.5 the right one?

Exercise 17.3. (The tilted boundary) Show directly on one site that $q\hat Z_q\to W'(\hat\Phi)$. Why does this not contradict Lemma 16.1, where the error Z of the untilted model has $qZ_q\to0$?

18 Identifying the limit

18.1 The deterministic coefficients

The coefficients M_q^N, $h_q^{[j],N}$, $H_q^N(y)$, $b_q^N(y)$ are constructed at each lattice. Their uniform bounds (Proposition 11.1) do not by themselves identify a limit: a weakly convergent vertex cannot in general be contracted against a varying covariance. The three tools are:

- *Weak vertex measures.* Derivative kernels with bounded rooted norms are signed measures with bounded mass; they converge weakly along subsequences, and the centred kernel Θ_q converges in a weak topology controlled by its first moment. The limiting coefficients solve the same fixed-point problem, whose solution is unique in its ball; so the whole sequence converges.

- *Compact sandwich.* If $G_N \to G$ in Hilbert–Schmidt norm, and B_N are symmetric, uniformly bounded and converge weakly to B, with $\sup_N \|G_N \mathsf{B}_N\| \le \theta < 1$ and $\|G\mathsf{B}\| \le \theta$, then $(I + G_N \mathsf{B}_N)^{-1} G_N \to (I + G\mathsf{B})^{-1} G$ in Hilbert–Schmidt norm. The free resolvent is the compact factor that turns weak convergence of the Hessian into strong convergence of the response.
- *Contraction against a moving covariance.* The rows of the inverses are Hölder (any exponent $a < \frac{1}{2}$); with the Hilbert–Schmidt convergence given by the sandwich, the squared kernels $(H_q^N)^2$ converge uniformly to a continuous kernel. Weakly convergent vertices of bounded variation can therefore be contracted against them, and the trace steps of the recursion pass to the limit.

With these, $H_q^N(Y_q^N) \to H_q(Y_q)$ and $b_q^N(Y_q^N) \to b_q(Y_q)$ in $L^2(I;\mathrm{HS})$ and $L^2(I;H)$ along strongly converging forecasts with the uniform supremum moments of Lemma 17.1.

18.2 The noise

The delicate point is the backward martingale. Fields converging in law do not imply that the stochastic integrals converge to stochastic integrals of the limits.

Remark 18.1. (Hidden noise) Let B be a real Brownian motion and $L_t^n = \int_0^t (1 + \sin(ns))\,\mathrm{d}B_s$. Then (L^n, B) converges in law to $(B + 2^{-1/2}\tilde{B}, B)$, with $\tilde{B}$ an independent Brownian motion: the integrands converge weakly to 1, but their squares converge weakly to $\frac{3}{2}$. A limiting FBSDE obtained this way would have an extra martingale, orthogonal to the driving noise.

To exclude this one needs *strong* compactness of the integrands. For the forward equation the integrand $H(Y)$ is an explicit function of Y, and strong convergence follows from the coefficient limits. For the backward equation the integrand is $\Xi = \mathrm{Cov} - H(Y)$, and we need strong compactness of the posterior covariance Cov_q. It comes from its exact dynamics.

Lemma 18.2. (Riccati equation of the posterior covariance) *At each N, $\mathrm{Cov}_q = D^2\mathscr{K}_q(P_q) \ge 0$ and*

$$\mathrm{d}\mathrm{Cov}_q = -\mathrm{Cov}_q^2\,\mathrm{d}q + D^3\mathscr{K}_q(P_q)[\mathrm{d}W_q]. \tag{18.1}$$

Proof. Differentiate the equation of Proposition 10.3(c) twice in p: $(\partial_q + \frac{1}{2}\Delta_p + D\mathscr{K}\cdot D_p)\,D^2\mathscr{K} = -(D^2\mathscr{K})^2$, and apply Itô's formula along $\mathrm{d}P = D\mathscr{K}(P)\mathrm{d}q + \mathrm{d}W$. □

This is (8.9) for $\partial_q \mathscr{R} = 1$: conditional variance is resolved at the rate Cov^2. For a smooth test f, the process $\langle f, \mathrm{Cov}_q f\rangle \ge 0$ is a nonnegative supermartingale, with drift $-\|\mathrm{Cov}_q f\|^2$, and initial value $\mathrm{Var}_{\nu_N}\langle \Phi_\infty, f\rangle$, bounded uniformly in N by Corollary 17.3. A nonnegative supermartingale has controlled oscillations: after stopping at the first time it exceeds a level K (an event of probability at most C_f/K), its decreasing part A and the bracket of its martingale part J satisfy $\mathbb{E}\,A_Q \le C_f$ and $\mathbb{E}[J]_Q \le K^2 + 2KC_f$, which gives time regularity in $W^{a,1}$, $a < \frac{1}{2}$, and hence strong compactness of each finite block $\langle f_i, \mathrm{Cov}\, f_j\rangle$. Spatial tails come from the isometry (recall $\psi_{q_0} = \mathbb{E}\,\Phi_\infty = 0$, ν_N being even), with symmetry exchanging input and output tails, $\mathbb{E}\int_I \|(I - \mathsf{F}_m)\mathrm{Cov}_q\|_{\mathrm{HS}}^2 \mathrm{d}q = \mathbb{E}\|(I - \mathsf{F}_m)\,\psi_Q\|_2^2$, and from the terminal averaging $\psi_Q = \mathbb{E}_Q \int_L^{L+1} (Y_s + Z_s)\mathrm{d}s$, using the H^σ bound of Proposition 17.2 and $\mathbb{E}\|Z_s\|_2^2 \le C\,s^{-5/2}$ (first $m \to \infty$, then $L \to \infty$). Altogether Cov^N is tight in strong $L^2(I;\mathrm{HS})$. Since $S \mapsto \int S\,S^*$ is continuous from $L^2(I;\mathrm{HS})$ to $C(I;\text{trace class})$, the brackets of ψ^N are tight in $C(I)$; a continuous martingale with tight brackets and uniformly small maximal Fourier tails (from the terminal averaging above and Doob's inequality) is tight in $C(I;H)$. Hence ψ^N, and $Z^N = \psi^N - Y^N$, are tight in $C(I;H)$. Strong convergence of the integrands gives convergence of the brackets $\int \mathrm{Cov}\mathrm{Cov}^*$ and of the covariations with W, and then:

Proposition 18.3. (Identification of the martingales [FBSDE26]) *Along a subsequence, on a common filtered probability space, the limit of the forecast satisfies $\psi_q = \int_{q_0}^{q} \mathrm{Cov}_s \, \mathrm{d}W_s$ with W a cylindrical Brownian motion for the joint filtration, and Y, Z, X satisfy*

$$Y_q = \int_{q_0}^{q} [b_s(Y_s) + H_s(Y_s)\, Z_s]\mathrm{d}s + \int_{q_0}^{q} H_s(Y_s)\, \mathrm{d}W_s,$$
$$Z_q = -\int_{q_0}^{q} [b_s(Y_s) + H_s(Y_s)\, Z_s]\mathrm{d}s + \int_{q_0}^{q} \Xi_s \, \mathrm{d}W_s, \qquad X_q = \int_{q_0}^{q} C_s \, \mathrm{d}W_s. \tag{18.2}$$

The argument is: the limiting ψ is a continuous square-integrable martingale for the joint filtration (fixed-time martingale identities pass to the limit with uniform integrability); its bracket and its covariation with W are those of $\int \mathrm{Cov}\, \mathrm{d}W$; so $\psi - \int \mathrm{Cov}\, \mathrm{d}W$ has zero bracket and vanishes. No extra martingale survives. The same argument for the forward equation identifies its noise with the same W, and identifies the limit of the auxiliary Gaussian field X as $\int C\, \mathrm{d}W$, not a new Gaussian field.

18.3 The result

Theorem 18.4. (Subsequential construction of Φ^4_3 [FBSDE26]) *Let $\lambda > 0$ and q_0 large enough. Along a subsequence of lattices:*

a) the embedded processes (Y^N, Z^N, ψ^N, X^N) (with $X^N_q = \int_{q_0}^{q} C_s \, \mathrm{d}W^N_s$ the auxiliary Gaussian field) converge in law in $C(I; H)$ for every finite I, together with the embedded innovations W^N in $C(I; H^{-r})$, $r > \frac{3}{2}$, and the integrands (Cov^N, Ξ^N) in strong $L^2(I; \mathrm{HS})$, jointly and with an almost sure representation;

b) the limit solves the FBSDE (18.2) on $[q_0, \infty)$, with $Y_{q_0} = Z_{q_0} = 0$, the deterministic continuum coefficients b, H, limits of those of Lecture 5 and unique in their ball, an H-cylindrical Brownian motion W for the joint filtration, obtained as the limit of the lattice innovations, a symmetric nonnegative integrand $\mathrm{Cov} = H(Y) + \Xi$, and $\mathbb{E}\|Z_q\|_2^2 \le C q^{-5/2}$;

c) there is a terminal field Φ with $Y_q \to \Phi$ in H^{-s}, almost surely and in mean square, for every $s > \frac{1}{2}$. Its law ν is the weak limit of the lattice measures ν_N along the subsequence, and for smooth h,

$$\langle Y_q + Z_q, h\rangle = \mathbb{E}[\langle \Phi, h\rangle \mid \mathcal{F}_q]: \tag{18.3}$$

the forecast remains the conditional mean of the terminal field;

d) the limiting thresholded energies E_∞, J_∞ satisfy the bounds of Theorem 16.2. Indeed, the lattice identities $V_q = C_q\,[(q - q_0)\,\psi_q - \frac{\gamma}{3} Y_q^3]$ and $\zeta_q = X_q + C_q \int_{q_0}^{q} (\psi_s - X_s)\mathrm{d}s - (q - q_0)\, C_q\, \psi_q$ need neither ϱ nor W, so V^N and ζ^N converge by (a), the L^6 convergence of Y^N and the Hilbert–Schmidt convergence of the free resolvent; the integrands of E and J are lower semicontinuous in these variables, and Fatou's lemma applies.

So the posterior structure survives the continuum limit: $\psi = Y + Z$ is a closed observable, the conditional mean of the terminal field; Y is computed from the explicit action; and the error Z is small at high resolution. This is the FBSDE of Lecture 4, in forecast coordinates, for a subsequential limit ν of the renormalised lattice measures, with the full noise retained.

18.4 The renormalised field equation

The terminal law satisfies an integration by parts formula in which the renormalised cube appears only tested in expectation.

Theorem 18.5. (Integration by parts) *Let $h, h_1, \dots, h_k$ be smooth and $G(\varphi) = g(\langle\varphi, h_1\rangle, \dots, \langle\varphi, h_k\rangle)$ with $g \in C_b^\infty(\mathbb{R}^k)$, and let $D_h G(\varphi) = \frac{\mathrm{d}}{\mathrm{d}a} G(\varphi + a\,h)|_{a=0}$. Along the subsequence of Theorem 18.4 (recall $C = (m^2 - \Delta_N)^{-1}$ with $m^2 = q_0$),*

$$\lim_N \mathbb{E}\,[G(\iota_N \Phi_\infty^N)\,\langle \mu_N \Phi_\infty^N + \lambda (\Phi_\infty^N)^3, j_N h\rangle_{H_N}] = \mathbb{E}_\nu\,[D_h G(\Phi) - G(\Phi)\langle \Phi, (q_0 - \Delta)\,h\rangle]. \tag{18.4}$$

Proof. At fixed N, Gaussian integration by parts in the density $\mathrm{e}^{-V_N - \frac{1}{2}\langle\varphi, K_{q_0}\varphi\rangle}$ gives, with $x = \Phi_\infty^N$ and $h_N = j_N h$,

$$\mathbb{E}\,[G(\iota_N x)\langle \mu_N x + \lambda x^3, h_N\rangle] = \mathbb{E}\,[D_{h_N}(G \circ \iota_N)(x)] - \mathbb{E}\,[G(\iota_N x)\langle x, K_{q_0} h_N\rangle].$$

Cell averaging commutes with lattice translations, so $K_{q_0} j_N h = j_N (q_0 - \Delta_N^{\mathrm{tr}})\,h$ with Δ_N^{tr} the finite-difference Laplacian acting on smooth functions, which converges to Δh in every Sobolev norm. The right side converges by Corollary 17.3 (weak convergence in H^{-s} and uniform moments), and the derivative term is a finite sum with converging coefficients $\langle j_N h, j_N h_i\rangle \to \langle h, h_i\rangle$. □

This is the terminal case of the Dyson–Schwinger equation of Theorem B.1, (B.3) of Lecture 4, for the limiting theory. It does not say that $\mu_N \Phi + \lambda \Phi^3$ converges as a random distribution: only its expectation against cylinder functions is identified. This is consistent with the expectation of Lecture 4 (Figure 7.1 and the remark following it) that the limiting force is an observable without a terminal representative; it is not proved here that no such representative exists. At every *finite* resolution, on the other hand, the force is explicit: $F_q = K_q Z_q - D\bar{\Gamma}_q(Y_q)$ (Exercises 16.1 and 18.1).

Theorem 18.5 is the statement at infinite resolution, where the cube can only be tested in expectation. At every finite resolution there is a statement between observables, in which no divergent quantity appears. In the limit of Theorem 18.4, let

$$P_q = W_q + \int_{q_0}^q \psi_s\,\mathrm{d}s, \quad \Phi_q = C_q P_q, \quad \eta_q = P_q - (q - q_0)\,\psi_q, \quad \langle F_q, h\rangle = \langle \psi_q, K_q h\rangle - \langle P_q, h\rangle,$$

and $J_q = (q - q_0)\,C_q$, which is $K_{q_0} Q_q$ in the notation of Lecture 4 and tends to I as $q \to \infty$. Here $\Phi_q \in H$, while F_q and η_q are distributions, tested with smooth h.

Theorem 18.6. (The field equation at resolution q) *Along the subsequence of Theorem 18.4, for every $q \geq q_0$ and all smooth h, G as in Theorem 18.5:*

a) Field equation. *Almost surely, as distributions,*

$$K_{q_0} \Phi_q = J_q F_q + \eta_q. \tag{18.5}$$

b) Observables. *$\langle F, h\rangle$ and $\langle \eta, h\rangle$ are centred martingales,*

$$\langle F_q, h\rangle = \int_{q_0}^q \langle (\mathrm{Cov}_s K_s - I)\,h, \mathrm{d}W_s\rangle, \qquad \langle \eta_q, h\rangle = \int_{q_0}^q \langle (I - (s - q_0)\,\mathrm{Cov}_s)\,h, \mathrm{d}W_s\rangle,$$

and $\mathbb{E}\,\mathrm{e}^{\langle \eta_q, h\rangle} \leq \mathrm{e}^{(q - q_0)\|h\|_2^2/2}$.

c) Dyson–Schwinger equation at resolution q.

$$\mathbb{E}\,D_{J_q h} G(\Phi_q) = \mathbb{E}[G(\Phi_q)\langle \eta_q, h\rangle] = \mathbb{E}[G(\Phi_q)(\langle \Phi_q, K_{q_0} h\rangle - \langle F_q, J_q h\rangle)]. \tag{18.6}$$

d) The renormalised cube as an observable. *With $\langle D\bar{\Gamma}_q(Y_q), h\rangle := \langle P_q, h\rangle - \langle Y_q, K_q h\rangle$, the explicit force of Lecture 5,*

$$\|\langle F_q, h\rangle + \langle D\bar{\Gamma}_q(Y_q), h\rangle\|_{L^2(\mathbb{P})} \leq C_\gamma \|h\|_{H^2} q^{-1/4}. \tag{18.7}$$

So $\langle F, h\rangle$ is the unique martingale asymptotically equivalent to the explicit process $-\langle D\bar{\Gamma}_q(Y_q), h\rangle$, in the sense of Lemma 5.4 of Lecture 3.

e) Infinite resolution. *As $q\to\infty$, both sides of* (18.6) *converge, and*

$$\lim_{q\to\infty}\mathbb{E}[G(\Phi_q)\langle D\bar\Gamma_q(Y_q),J_qh\rangle]=\mathbb{E}_\nu[D_hG(\Phi)-G(\Phi)\langle\Phi,(q_0-\Delta)\,h\rangle], \tag{18.8}$$

the right side of (18.4).

Proof. (a) is algebra: $J_qK_q=q-q_0$ and $K_{q_0}=K_q-(q-q_0)$ give, for $\langle\Phi_q,K_qk\rangle=\langle P_q,k\rangle$, the identity $K_{q_0}\Phi_q-J_qF_q=P_q-(q-q_0)\,\Phi_q-(q-q_0)\,(\psi_q-\Phi_q)=\eta_q$.

(c) *At fixed N.* By (13.1), $\eta_q=\mathbb{E}_q[B_{q-q_0}]$, and $G(\Phi_q)$ is $\mathcal{F}_q$-measurable, so $\mathbb{E}[G(\Phi_q)\langle\eta_q,h\rangle]=\mathbb{E}[G(\Phi_q)\langle B_{q-q_0},h\rangle]$. Now $\Phi_q=C_qP_q=J_q\Phi_\infty+C_qB_{q-q_0}$, with B independent of Φ_∞. Gaussian integration by parts in B gives $(q-q_0)\,\mathbb{E}\,D_{C_qh}G(\Phi_q)=\mathbb{E}\,D_{J_qh}G(\Phi_q)$. *In the limit.* Apply this with j_Nh. By Theorem 18.4(a), $\iota_N\Phi_q^N\to\Phi_q$ in H: from $K_qX_q=W_q+\int_{q_0}^q X_s\,\mathrm{d}s$ we get $\Phi_q^N=X_q^N+C_q^N\int_{q_0}^q(\psi_s^N-X_s^N)\,\mathrm{d}s$, where X^N and ψ^N converge in $C(I;H)$ and $\iota_NC_q^Nj_N\to C_q$ in operator norm on H; $\langle\eta_q^N,j_Nh\rangle$ converges in L^2 because it involves only W^N and ψ^N tested with h; and $\iota_NJ_q^Nj_Nh\to J_qh$. The moments are uniform in N. The second equality in (18.6) is (a).

(b) By Proposition 18.3, $\psi_q=\int\mathrm{Cov}_s\,\mathrm{d}W_s$; insert $P=W+\int\psi$ and use the product rule, the drifts cancelling: $\mathrm{d}(K_q\psi_q-P_q)=(K_q\mathrm{Cov}_q-I)\,\mathrm{d}W_q$, tested with h, and similarly for η. The exponential bound holds at fixed N by Lemma 13.2, and passes to the limit by Fatou.

(d) At fixed N, $P_q=K_qY_q+D\bar\Gamma_q(Y_q)$ and $F_q=K_qZ_q-D\bar\Gamma_q(Y_q)$. So the left side of (18.7) is $\|\langle Z_q,K_qh\rangle\|_{L^2}\le\|K_qh\|_2\,(\mathbb{E}\|Z_q\|_2^2)^{1/2}\le C_\gamma\|h\|_{H^2}q^{-1/4}$, by Theorem 16.2. The bound passes to the limit.

(e) By (18.7) with the exact $K_qJ_qh=(q-q_0)\,h$, replacing $-F_q$ by $D\bar\Gamma_q(Y_q)$ in (18.6) costs $O(q^{-1/4})$. Moreover $\langle\Phi_q,k\rangle=\langle\psi_q,J_qk\rangle+\langle\zeta_q,k\rangle$, where $\mathbb{E}\langle\zeta_q,k\rangle^2\le(q-q_0)\|C_qk\|_2^2\le\|k\|_2^2/q$, and $\psi_q\to\Phi$ in H^{-s} in mean square by Theorem 18.4(c). So $\Phi_q\to\Phi$ tested with smooth functions, and $J_qh\to h$ in every Sobolev norm. □

So the field equation holds at every resolution with no renormalisation constant, and in the strong sense of a pathwise identity between distributions. The nonlinearity enters only through the force observable F, whose class contains the explicit process $-D\bar\Gamma_q(Y_q)$: the cube of the forecast with its counterterms, at resolution q. This is the renormalised cube, as an element of the space of observables of Lecture 3. Theorem 18.5 is the shadow of this at infinite resolution: by (18.8), the two orders of limits, $N\to\infty$ at the terminal resolution and $q\to\infty$ after N, give the same right side. What is not proved is that F is closed: the statement (18.7) gives no bound on $\langle F_q,h\rangle$ uniform in q.

18.5 Dyson–Schwinger along the scales

Theorem 18.6 is stated for functionals of the field at one resolution. The same proof works along the whole observed path up to q, and this is the scale-by-scale form of the Dyson–Schwinger equations of Lecture 4 for the interacting theory.

Corollary 18.7. (Dyson–Schwinger along the scales) *Along the subsequence of Theorem 18.4, let $q_0\le q_1\le\ldots\le q_k\le q$, let $h,h_1,\ldots,h_k$ be smooth, $g\in C_b^\infty(\mathbb{R}^k)$ and $G=g(\langle\Phi_{q_1},h_1\rangle,\ldots,\langle\Phi_{q_k},h_k\rangle)$. Then*

$$\mathbb{E}[G\langle\eta_q,h\rangle]=\mathbb{E}\,\mathcal{D}_hG,\quad \mathcal{D}_hG=\sum_{i=1}^k\partial_ig\langle J_{q_i}h,h_i\rangle, \tag{18.9}$$

that is, the derivative of G in the direction which shifts the field at each scale s by J_sh. Equivalently, for $\langle\eta_q,h\rangle=\langle\Phi_q,K_{q_0}h\rangle-\langle F_q,J_qh\rangle$,

$$\mathbb{E}\,\mathcal{D}_hG=\mathbb{E}[G(\langle\Phi_q,K_{q_0}h\rangle+\langle D\bar\Gamma_q(Y_q),J_qh\rangle)]+O(C_\gamma\|g\|_\infty\|h\|_{H^2}q^{-1/4}). \tag{18.10}$$

So at every resolution the renormalised cube $D\bar\Gamma_q(Y_q)$ closes the hierarchy, up to an error which vanishes as $q\to\infty$. The increments are consistent: $\mathbb{E}[G\langle\eta_q-\eta_s,h\rangle]=0$ for $q_k\le s\le q$.

Proof. At fixed N, G is $\mathcal{F}_q$-measurable, so $\mathbb{E}[G\langle\eta_q,h\rangle]=\mathbb{E}[G\langle B_{q-q_0},h\rangle]$ by (13.1). Moreover $\Phi_s=J_s\Phi_\infty+C_sB_{s-q_0}$ with B independent of Φ_∞, and for $s\le q$ the covariance of $\langle B_{q-q_0},h\rangle$ and $\langle C_sB_{s-q_0},k\rangle$ is $(s-q_0)\langle C_sh,k\rangle=\langle J_sh,k\rangle$. Gaussian integration by parts in the path of B gives (18.9). The limit is taken as in the proof of Theorem 18.6, now using the joint convergence at the scales $q_1,\dots,q_k,q$. Finally (18.10) follows from (18.7) with J_qh, since $\|J_qh\|_{H^2}\le\|h\|_{H^2}$. □

18.6 Uniqueness in finite volume

The construction above is subsequential, and nothing in the FBSDE selects the limit. In finite volume uniqueness can be obtained from outside, as in [HSt22], by identifying ν as the invariant measure of the Φ^4_3 dynamics. The measure ν_N is invariant for the lattice equation

$$\mathrm{d}\varphi=[(\Delta_N-q_0-\mu_N)\varphi-\lambda\varphi^3]\,\mathrm{d}t+\sqrt{2}\,\mathrm{d}W^N_t, \tag{18.11}$$

with W^N a Brownian motion on H_N. Write $\mu_N=m_N+c_N$, where m_N is a choice of the renormalisation constants of the lattice dynamics in [HMt18]. With $\ell=1/(2\lambda)$, the change of variables $\varphi(t,x)=\sqrt{2}\ell^{-1/2}v(t/\ell^2,x/\ell)$ gives unit cube and unit noise on a torus of side 2λ, and q_0 goes into the finite part. We assume the following three facts in this setting, with $\alpha=-\frac{1}{2}-\kappa$.

i. **Convergence of the lattice dynamics.** If $c_N\to c$, the solutions of (18.11) converge in probability to the solution of the Φ^4_3 equation with parameter c, uniformly over initial conditions in compact subsets of C^α.

ii. **Ergodicity.** The Φ^4_3 semigroup $\mathcal{P}^c_t$ has a unique invariant probability measure π_c, and $\|\mathcal{P}^c_t\mu-\pi_c\|_{\mathrm{TV}}\to0$ uniformly in the initial law μ.

iii. **Tightness in C^α.** For bounded c_N the lattice measures are tight in C^α. Corollary 17.3 gives only H^{-s}.

These are not quite the statements in the literature, and this must be kept in mind. Facts (i) and (iii) are proved in [HMt18, HSt22] and [GH21] for dyadic grids, with a fixed finite part, while our N is odd and c_N may vary. Fact (ii) follows from the strong Feller property [HMa18], the support theorem [HSc22] and coming down from infinity [MW17, MW20], which are stated on the unit torus with unit coefficients for the cube and the noise. We expect these extensions to be routine, but we have not checked them.

Theorem 18.8. (Uniqueness in finite volume, conditional) *Assume (i)–(iii). If $c_N\to c$ along a subsequence, then $\iota_N\nu_N\to\pi_c$ along it. Consequently, if c_N converges, the whole sequence $\iota_N\nu_N$ converges, and $\nu=\pi_c$ does not depend on the subsequence. Moreover, in the limit of Theorem 18.4, $P_q=(q-q_0)\Phi+B_{q-q_0}$ with B a Brownian motion independent of Φ, so the joint law of (Φ,P) is unique.*

Proof. The argument is that of [HSt22], Theorem 2.2. Let f be bounded and Lipschitz on C^α. By invariance of ν_N under the lattice semigroup $\mathcal{P}^N_t$,

$$|\pi_c(f)-\iota_N\nu_N(f)|\le|\pi_c(f)-\mathcal{P}^c_t(\iota_N\nu_N)(f)|+|\mathcal{P}^c_t(\iota_N\nu_N)(f)-\iota_N(\mathcal{P}^N_t\nu_N)(f)|.$$

By (ii) the first term is small for t large, uniformly in N. For that fixed t, split the second term on a compact set of initial conditions with almost full mass, by (iii), and use (i) there. For the last claim, $B^N=P^N-(q-q_0)\Phi^N_\infty$ converges jointly with Φ^N_∞ by (17.4), and independence passes to weak limits. □

Corollary 18.9. (Quartic tails) *If $\nu=\pi_c$, then $\mathbb{E}_\nu\exp(\beta\langle\Phi,\psi\rangle^4)<\infty$ for smooth ψ and small $\beta>0$. This is [HSt22], Theorem 1.1, proved there in fixed volume and for any fixed torus size. The homogeneous version of [HSt22], Theorem 1.2, stated there as a direct adaptation, would then give $\mathbb{E}_\nu\exp(\beta\|\Phi\|^4_{H^{-1/2-\kappa}})<\infty$, adding the zero mode through Theorem 1.1 and a partition of unity; that version is stated there only on the unit torus, so on the side 2λ this is a further assumed extension.*

Three things should be kept apart. **First**, this proof uses none of the estimates of this lecture: it only identifies the object constructed here, under the assumptions above. **Second**, neither the convergence nor the boundedness of c_N has been proved here, and identifying a limit point as some π_c needs at least boundedness along a subsequence. This is a statement about lattice sums. By (11.9) of Lecture 5, $\mu_N = \int_{q_0}^{\infty} \beta_q \, \mathrm{d}q$, and power counting predicts that, after the tadpole and the sunset are removed, the remainder is integrable at $q = \infty$ uniformly in N. The removal is not term by term, since the double-tadpole logarithms cancel only between the two terms of (11.8), and the coefficient of q^{-1} must match the lattice sunset exactly. **Third**, the theorem does not say that the limiting FBSDE has a unique solution as an equation. To deduce uniqueness of the law of (Y, Z, W) one would also want $\psi_q = \mathbb{E}[\Phi | \mathcal{F}_q^P]$ and Y_q a function of P_q in the limit, which is not shown here. An intrinsic proof in the language of this lecture remains open.

18.7 What is not proved, and where to go next

- *Uniqueness.* In finite volume Theorem 18.8 identifies limit points through the dynamics, under explicit assumptions: extensions of known results to our lattices and the convergence of the finite part c_N. Not proved here: these assumptions, uniqueness for the limiting FBSDE as an equation, and an intrinsic proof.

- *Small mass* (*strong coupling*). The construction needs $\gamma \, q_0^{-1/2}$ small. By scaling, it is equivalent to the weak-coupling regime $\lambda / m \ll 1$ (unit mass on a torus of side m). Removing it requires a priori estimates that do not rely on the smallness of the feedback coefficient γ^2 / q_0, and a coefficient construction outside the perturbative ball ($g_{q_0} = \gamma \, q_0^{-1/2}$ small in Proposition 11.1).

- *Axioms.* Reflection positivity holds at every lattice, for reflections in lattice and mid-lattice planes (nearest-neighbour action), and passes to ν for every reflection in a coordinate plane, by approximating the plane with lattice planes (standard; Exercise 18.4). Rotation invariance (absent on the torus) and clustering need infinite volume.

- *Infinite volume.* The force, its Burgers flow, and differences of free energies are intensive (Section 9.2). The estimates of this lecture must then be made local, with weights in space and constants per unit volume. A new difficulty appears in the backward equation: its integrand Ξ is a nonlocal operator, and the backward error at a point can receive noise from arbitrarily far away. Uniform-in-volume coefficient bounds and local energy moments are themselves open; they could give an infinite-volume Gibbs subsequence. Controlling the "noise from infinity" (together with local compactness of the Hessian blocks) is the main additional step needed to retain the full FBSDE without an orthogonal martingale [FBSDE26].

Summary of the course. The six lectures went from the Wightman axioms to the Euclidean formulation (Lecture 1), to the free field (Lecture 2), to stochastic quantisation and the scale decomposition (Lecture 3), and to the three faces of an interacting field: the Polchinski equation, the FBSDE and the variational problem (Lecture 4). In Lectures 5–6 the FBSDE in forecast coordinates, together with an explicit solution of the Wetterich equation up to an integrable defect, gave a construction of Φ_3^4 in which every object has a probabilistic meaning: the forecast is a conditional mean, the response is a conditional covariance, the Brownian motion is an innovation, and UV stability is the statement that an explicit action predicts the conditional mean up to an error that is small at high resolution.

Lecture 4 (formal)	Lectures 5–6	status in the limit
forecast $\psi_t = \mathbb{E}_t \Phi_T$	$\psi = Y + Z$	closed martingale, (18.3)
response $H_t = \mathrm{Cov}_t$	$\mathrm{Cov} = H(Y) + \Xi$	strong $L^2(\mathrm{HS})$, Riccati
Γ_t, Wetterich	$\bar\Gamma_q$ explicit, defect e	deterministic coefficients
force F_t	$K_q Z_q - D\bar\Gamma_q(Y_q)$	observable at finite q
DS equation	Theorem 18.5	tested in expectation

Exercises

Exercise 18.1. (**The force observable**) Show that $\langle F_q^N, j_N h\rangle = \langle \psi_q^N, K_q j_N h\rangle - \langle W_q^N, j_N h\rangle - \int_{q_0}^q \langle \psi_s^N, j_N h\rangle \mathrm{d}s$. Deduce that, along the subsequence of Theorem 18.4, these converge, with uniformly bounded moments, to $\langle F_q, h\rangle = \int_{q_0}^q \langle (\mathrm{Cov}_s K_s - I)\, h, \mathrm{d}W_s\rangle$, a martingale for the limiting filtration (note $F_{q_0} = 0$, since ν_N is even). (Cross-check: $F_q = K_q Z_q - D\bar\Gamma_q(Y_q)$, with Lemma 17.1 and Theorem 16.2, gives a uniform second-moment bound.) Why does this not give a terminal value F_∞?

Exercise 18.2. (**Hidden noise**) Prove the claim of Remark 18.1. Where would the argument of Proposition 18.3 fail if only weak convergence of Cov^N were known?

Exercise 18.3. (**Riccati and positivity**) For a Gaussian $\nu_N = \mu^C$, solve (18.1) and check that $\mathrm{Cov}_q = C_q$. For a general ν_N, show that TrCov_q is a supermartingale.

Exercise 18.4. (**Reflection positivity**) Show that ν_N is reflection positive for reflections in lattice and mid-lattice planes (nearest-neighbour Gaussian times a local potential). Deduce reflection positivity of ν for every coordinate plane $\{x_1 = a\}$: take cylinder functions of tests supported in $\{a < x_1 < a + \frac{1}{2}\}$ (one of the two components of the torus minus the fixed planes $\{x_1 = a\}$ and $\{x_1 = a + \frac{1}{2}\}$ of θ_a), cell-average them, and use lattice planes $a_N \to a$, so that for large N the averaged tests lie on the correct side. Since lattice and mid-lattice reflections map cells to cells, $\theta_{a_N} j_N f = j_N \theta_{a_N} f$, so the reflected tested variables are $\langle \iota_N \Phi, \theta_{a_N} f\rangle$, with $\theta_{a_N} f \to \theta_a f$ in every H^s.

Exercise 18.5. (**Dyson–Schwinger by Gaussian integration by parts**) (a) At fixed N, check that (18.6) is Theorem B.1 of Lecture 4 in the profile of Lecture 5, and compare the two proofs: score of the law of Φ_q against the filtered noise. (b) For $V_N = 0$, write $F = 0$ and check (18.5) directly. (c) Show that $\langle F, h\rangle$ has bracket $\int \|(\mathrm{Cov}_s K_s - I)\, h\|_2^2 \mathrm{d}s$, and that (18.7) does not bound it uniformly in q. What would a bound imply?

Exercise 18.6. (**Negative Sobolev threshold**) Verify (17.3), and show that at $s = \frac{3}{2}$ the bound becomes $Q^{-1}\log(2+Q)$.

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